Practical - 15

Practical Title: Final Integration, Testing, Documentation, GitHub Submission, and Local Deployment

Problem definition: Integrate all modules into a single deployable StudentHub portal. Verify navigation, responsiveness, authentication, database operations, security checks, AJAX functionality, and documentation. Deploy locally using XAMPP/Apache virtual host and submit GitHub repository.

A. Theory:

● Are all pages properly linked and navigable?

Yes, all required pages are linked through the navigation menu, and each page opens correctly without broken links.

● Are sessions, roles, validations, and database operations working end-to-end?

Yes, after testing, registration, login, logout, session management, role-based access, form validation, and database operations work correctly.

● Are security and accessibility checklists completed?

The application is checked for password hashing, prepared statements, session protection, input validation, image alt text, form labels, keyboard navigation, and readable contrast.

● Is project documentation sufficient for installation and testing?

Yes, the README explains the technologies, installation steps, database setup, features, and testing process. The project also includes an SQL dump and test report.

● Is the project repository clean and properly committed?

Yes, the required project files are committed and pushed to GitHub, and temporary files and sensitive credentials are excluded.

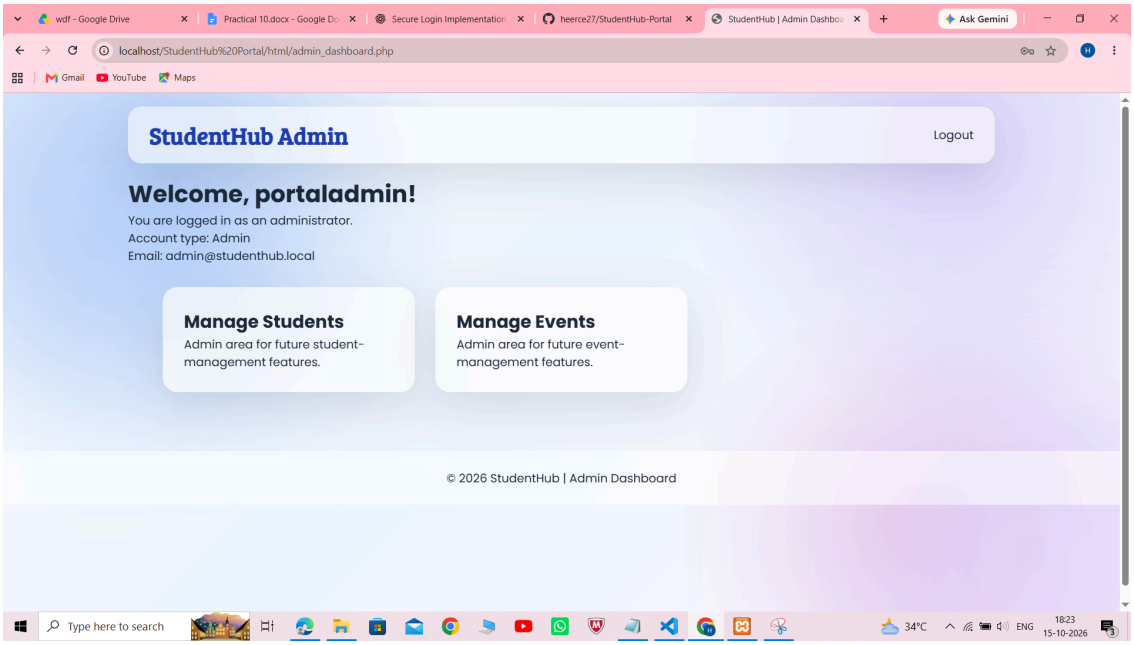
B. Implementation

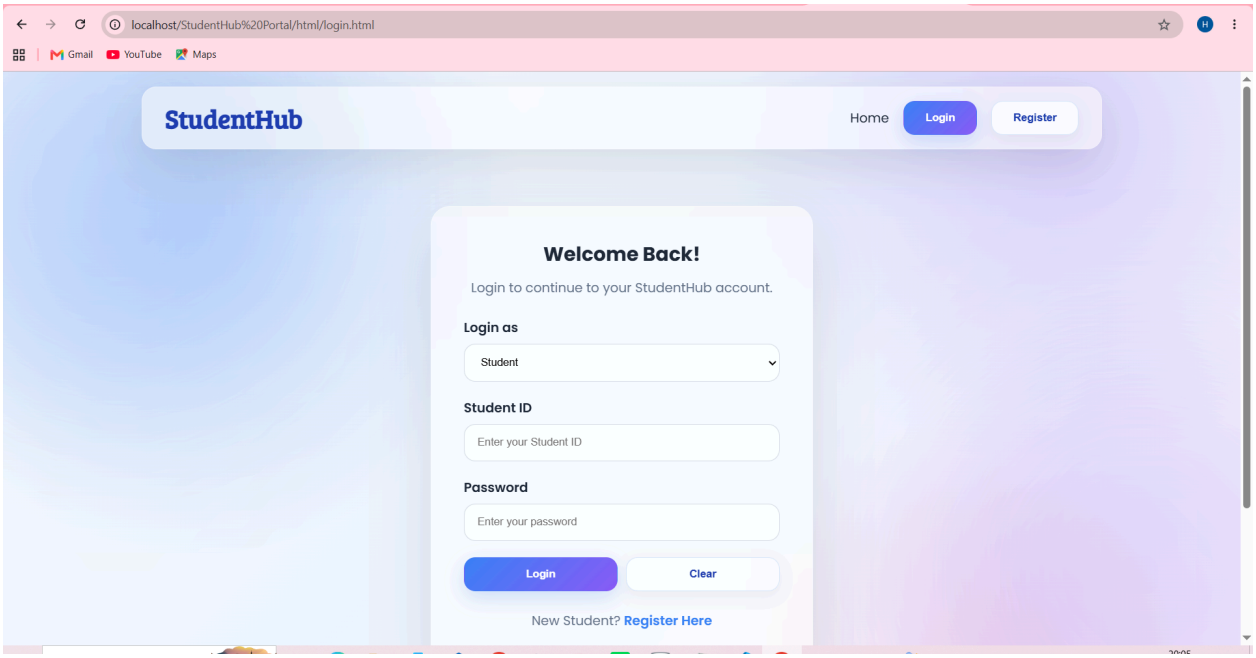
PHP syntax and functions used:

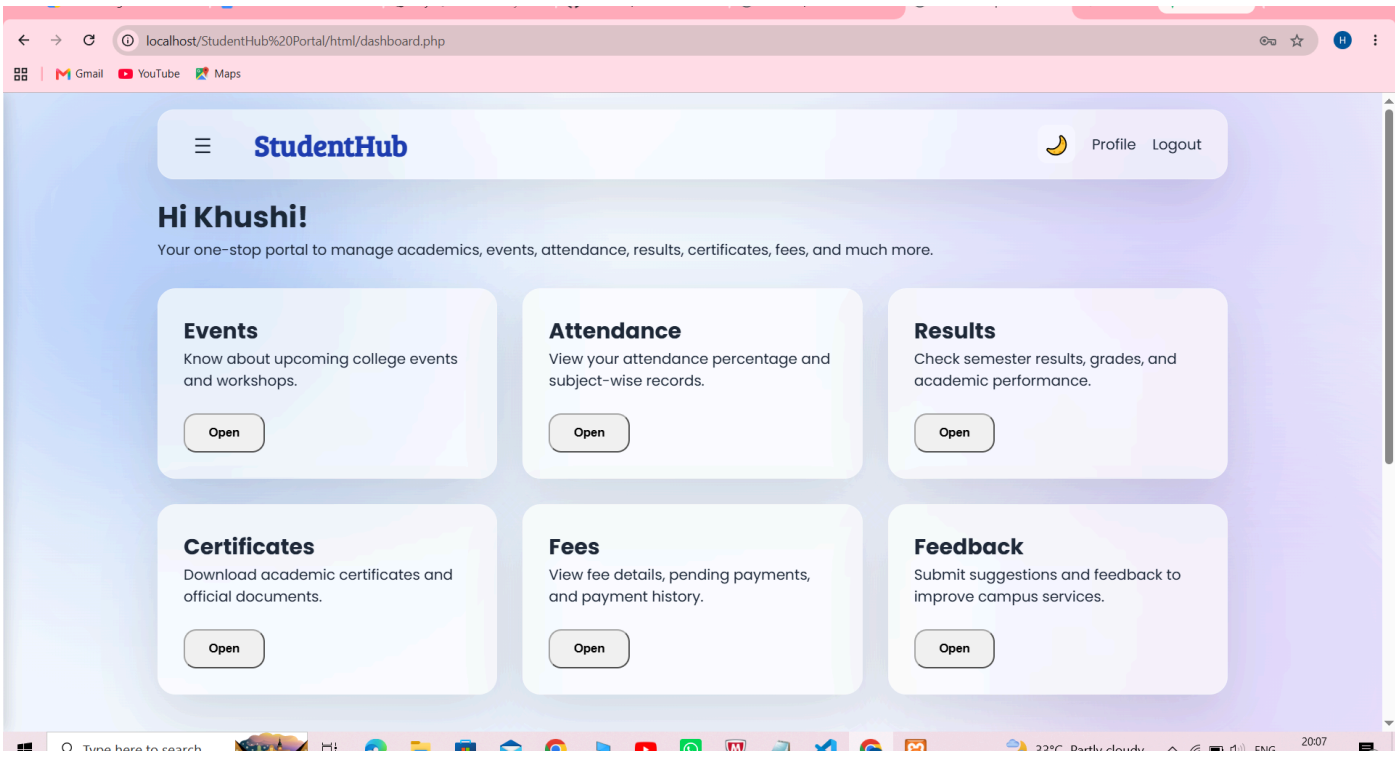
PHP element	Purpose

`require_once`	Includes the database connection file only once.
`new mysqli()`	Connects PHP to the MySQL database.
`if` and `else`	Check conditions and display appropriate output.
`while`	Repeats code to display each student record.
`fetch_assoc()`	Retrieves a database row as an associative array.
`htmlspecialchars()`	Helps prevent HTML injection when displaying database values.
`close()`	Closes the database connection.
`$_SERVER`	Provides server and request information when required.

C. Evidence of Work




localhost/StudentHub%20Portal/html/login.html
Gmail
YouTube
Maps
StudentHub
Home
Login
Register
Welcome Back!
Login to continue to your StudentHub account.
Login as
Student
Student ID
Enter your Student ID
Password
Enter your password
Login
Clear
New Student? Register Here


localhost/StudentHub%20Portal/html/dashboard.php
Gmail
YouTube
Maps
StudentHub
Profile
Logout
Hi Khushi!
Your one-stop portal to manage academics, events, attendance, results, certificates, fees, and much more.
Events
Know about upcoming college events and workshops.
Open
Attendance
View your attendance percentage and subject-wise records.
Open
Results
Check semester results, grades, and academic performance.
Open
Certificates
Download academic certificates and official documents.
Open
Fees
View fee details, pending payments, and payment history.
Open
Feedback
Submit suggestions and feedback to improve campus services.
Open

D. GitHub Repository Link:

https://github.com/heerce27/StudentHub-Portal.git

E. Challenges Faced:

- Login issues were fixed by verifying sessions and role checks.

Learning Outcomes:

After completing this practical, I learned how to integrate different web development modules into a single application.

Faculty Remarks: